

SOLIDWORKS API (.NET VBA C#) — Reference

(184/337): IModelDoc2 (part 7 of 17)

SOLIDWORKS API Documentation (converted from CHM)

IModelDoc2 (part 7 of 17) (Type)

GetDisplayWhenAdded Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetDisplayWhenAdded Method (IModelDoc2)

Gets whether new sketch entities are displayed when created.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDisplayWhenAdded() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Boolean  
  
value = instance.GetDisplayWhenAdded()
```

C#

```
System.bool GetDisplayWhenAdded()
```

C++/CLI

```
System.bool GetDisplayWhenAdded();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if new sketch entities are displayed when added, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetDisplayWhenAdded](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SetDisplayWhenAdded Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

GetEntityName Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetEntityName Method (IModelDoc2)

Gets the name of the specified face, edge, or vertex.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEntityName(  
    ByVal Entity As System.Object  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Entity As System.Object
Dim value As System.String

value = instance.GetEntityName(Entity)
```

C#

```
System.string GetEntityName(
    System.object Entity
)
```

C++/CLI

```
System.String^ GetEntityName(
    & System.Object^ Entity
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Entity
[Entity](#)

Return Value

Name of the entity

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetEntityName](#).

Example

[Select Component Face By Name \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IGetEntityName Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEquationMgr Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetEquationMgr Method (IModelDoc2)

Gets the equation manager.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEquationMgr() As EquationMgr
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As EquationMgr

value = instance.GetEquationMgr()
```

C#

```
EquationMgr GetEquationMgr()
```

C++/CLI

```
EquationMgr^ GetEquationMgr();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Equation manager](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetEquationMgr](#).

Example

[Use If Function When Adding an Equation \(VBA\)](#)
[Automatically Solve Equations in Sequence \(C#\)](#)
[Automatically Solve Equations in Sequence \(VB.NET\)](#)
[Automatically Solve Equations in Sequence \(VBA\)](#)
[Get Equation Values \(C#\)](#)
[Get Equation Values \(VB.NET\)](#)
[Get Equation Values \(VBA\)](#)

Remarks

Use the equation manager to work with the existing equations in a model and modify a specific equation.

The [IEquationMgr](#) object is associated with the configuration that was active at the time it was acquired. If you change the active configuration while holding an IEquationMgr object reference, release it and reacquire it if it is needed.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetExternalReferenceName Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetExternalReferenceName Method (IModelDoc2)

Gets the name of the externally referenced document (in the case of a join or mirrored part).

.NET Syntax

Visual Basic (Declaration)

```
Function GetExternalReferenceName() As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.String
```

```
value = instance.GetExternalReferenceName()
```

C#

```
System.string GetExternalReferenceName()
```

C++/CLI

```
System.String^ GetExternalReferenceName();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Full path of referenced document or NULL

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetExternalReferenceName](#).

Remarks

If the model document does not have an externally referenced document, a NULL string is returned.

This example shows how to get the names of any externally referenced document for the open model.

```
'-----
```

```
Option Explicit
```

```
Sub main()
```

```
    Dim swApp                As SldWorks.SldWorks
```

```
    Dim swModel              As SldWorks.ModelDoc2
```

```
    Set swApp = Application.SldWorks
```

```
    Set swModel = swApp.ActiveDoc
```

```
    Debug.Print "File = " & swModel.GetPathName
```

```
    Debug.Print "  ExtRefName = " & swModel.GetExternalReferenceName
```

```
End Sub
```

```
'-----
```

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::BreakAllExternalReferences Method](#)

[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)

[IModelDoc2::ListAuxiliaryExternalFileReferencesCount Method](#)

[IModelDoc2::LockAllExternalReferences Method](#)

[IModelDocExtension::ListExternalFileReferences Method](#)

[IModelDocExtension::ListExternalFileReferencesCount Method](#)

[IModelDocExtension::IListExternalFileReferences Method](#)

[IModelDocExtension::UpdateExternalFileReferences Method](#)

[IModelDoc2::IListAuxiliaryExternalFileReferences Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFeatureCount Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetFeatureCount Method (IModelDoc2)

Gets the number of features in this document.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFeatureCount() As System.Integer
```

```
Dim instance As IModelDoc2
Dim value As System.Integer

value = instance.GetFeatureCount()
```

C#

```
System.int GetFeatureCount()
```

C++/CLI

```
System.int GetFeatureCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of features in this document; this number does not include subfeatures

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetFeatureCount](#).

Example

[Traverse Features By Reverse Position \(C#\)](#)
[Traverse Features By Reverse Position \(VB.NET\)](#)
[Traverse Features By Reverse Position \(VBA\)](#)
[Get Features in Reverse Order \(C#\)](#)
[Get Features in Reverse Order \(VB.NET\)](#)
[Get Features in Reverse Order \(VBA\)](#)

Remarks

The number of features returned by this method does not include subfeatures. Subfeatures include, for example, mate features within a mategroup, drawing views on a sheet, and so on. One way to identify a subfeature is whether it can be returned by:

- [IFeature::GetFirstSubFeature](#) or [IFeature::!GetFirstSubFeature](#)
- [IFeature::GetNextSubFeature](#) or [IFeature::!GetNextSubFeature](#).

This method returns the number of features returned when traversing the Feature list with [IModelDoc2::FirstFeature](#) or [IModelDoc2::!FirstFeature](#) and [IFeature::GetNextFeature](#) or [IFeature::!GetNextFeature](#). This value may be useful in feature traversal or if accessing the feature by position using [IModelDoc2::FeatureByPositionReverse](#) or [IModelDoc2::!FeatureByPositionReverse](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFeatureManagerWidth Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetFeatureManagerWidth Method (IModelDoc2)

Gets the width of the FeatureManager design tree.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFeatureManagerWidth() As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim value As System.Integer

value = instance.GetFeatureManagerWidth()
```

C#

```
System.int GetFeatureManagerWidth()
```

C++/CLI

```
System.int GetFeatureManagerWidth();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Width

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetFeatureManagerWidth](#).

Example

[Change Width of FeatureManager Design Tree \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[SetFeatureManagerWidth Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFirstAnnotation Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetFirstAnnotation Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::GetFirstAnnotation2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetFirstAnnotation() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Object

value = instance.GetFirstAnnotation()
```

C#

```
System.object GetFirstAnnotation()
```

C++/CLI

```
System.Object^ GetFirstAnnotation();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetFirstAnnotation](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFirstAnnotation2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetFirstAnnotation2 Method (IModelDoc2)

Gets the first annotation in the model.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFirstAnnotation2() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Object  
  
value = instance.GetFirstAnnotation2()
```

C#

```
System.Object GetFirstAnnotation2()
```

C++/CLI

```
System.Object^ GetFirstAnnotation2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [annotation](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetFirstAnnotation2](#).

Example

[Get Display Dimensions, GTols, and Surface-finish Symbols \(VBA\)](#)
[Get DimXpert Display Dimensions and Feature \(C#\)](#)
[Get DimXpert Display Dimensions and Feature \(VB.NET\)](#)
[Get DimXpert Display Dimensions and Feature \(VBA\)](#)

Remarks

For parts and assemblies, this method returns the first [IAnnotation](#) object in the model. For drawings, access the annotations using the [IView::GetFirstAnnotation3](#) method.

A dimension becomes suppressed or hidden when you specifically select a dimension and hide it, or when you select a feature and you select to hide all dimensions. If you need to filter out these dimensions, use [IAnnotation::Visible](#) to check that status.

NOTE: The difference between the obsoleted [IModelDoc2::GetFirstAnnotation](#) and this method, [IModelDoc2::GetFirstAnnotation2](#), is that [IModelDoc2::GetFirstAnnotation2](#) retrieves any display dimension, including suppressed, hidden, or dangling dimensions.

To get the next annotation in the model, call [IAnnotation::GetNext3](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IGetFirstAnnotation2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFirstModelView Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetFirstModelView Method (IModelDoc2)

Gets the first view in a model document.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFirstModelView() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Object

value = instance.GetFirstModelView()
```

C#

```
System.Object GetFirstModelView()
```

C++/CLI

```
System.Object^ GetFirstModelView();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [model view](#) in this document

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetFirstModelView](#).

Example

[Run SOLIDWORKS Commands and Synthesize Mouse Events \(C#\)](#)
[Run SOLIDWORKS Commands and Synthesize Mouse Events \(VB.NET\)](#)
[Run SOLIDWORKS Commands and Synthesize Mouse Events \(VBA\)](#)

Remarks

You can traverse through the model views in a document by using this method to get the initial view and [IModelView::GetNext](#) to get each of subsequent views. You have reached the end of the list when IModelView::GetNext returns Nothing or null.

See [IModelDoc2::EnumModelViews](#) for traversing the model views on a document.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IGetFirstModelView Method](#)
[IModelDoc2::ActiveView Property](#)
[IModelDoc2::IActiveView Property](#)
[IModelDoc2::GetModelViewCount Method](#)
[IModelDoc2::GetModelViewNames Method](#)
[IModelDoc2::IGetModelViewNames Method](#)
[IModelDoc2::ModelViewManager Property](#)
[IModelDocExtension::GetModelViewCount Method](#)
[IModelDocExtension::GetModelViews Method](#)
[IModelDocExtension::IGetModelViews Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetGridSettings Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetGridSettings Method (IModelDoc2)

Gets the current grid settings.

.NET Syntax

Visual Basic (Declaration)

```
Function GetGridSettings() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Object
```

```
value = instance.GetGridSettings()
```

C#

```
System.Object GetGridSettings()
```

C++/CLI

```
System.Object^ GetGridSettings();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Current grid settings (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetGridSettings](#).

Remarks

The return format is the following array of doubles:

[*dispGrid*, *gridSpacing*, *snap*, *dotStyle*, *nMajor*, *nMinor*, *angleSnap*, *angleUnit*, *minorAuto*]

where:

- *dispGrid* - can be interpreted as a BOOLEAN: True if grid displayed, false otherwise
- *gridSpacing* - snap distance
- *snap* - can be interpreted as a BOOLEAN: True if snap to grid is on, false otherwise
- *dotStyle* - can be interpreted as a BOOLEAN: True if dotted grids, false otherwise
- *nMajor* - number of minors in major
- *nMinor* - number of snaps in minor
- *angleSnap* - can be interpreted as a BOOLEAN: True if snap to angle is on, false otherwise
- *angleUnit* - value of angle to which to snap
- *minorAuto* - can be interpreted as a BOOLEAN: True if the minor grids are set automatically, false otherwise

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GridOptions Method](#)
[IModelDoc2::ToolsGrid Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetInferenceMode Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetInferenceMode Method (IModelDoc2)

Obsolete. Superseded by [SketchManager::InferenceMode](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetInferenceMode() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Boolean
```

```
value = instance.GetInferenceMode()
```

C#

```
System.bool GetInferenceMode()
```

C++/CLI

```
System.bool GetInferenceMode();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetInferenceMode](#).

Remarks

This method affects sketch-entity snapping or inferring constraints to other geometry during creation.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AutoInferToggle Method](#)
[IModelDoc2::SetInferenceMode Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetLayerManager Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetLayerManager Method (IModelDoc2)

Gets the layer manager for the current drawing document.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLayerManager() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Object

value = instance.GetLayerManager()
```

C#

```
System.object GetLayerManager()
```

C++/CLI

```
System.Object^ GetLayerManager();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Layer manager](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetLayerManager](#).

Example

[Determine If Layer Is Visible \(VBA\)](#)
[Get Layers \(C#\)](#)
[Get Layers \(C#\)](#)
[Get Layers \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IGetLayerManager Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetLightSourceCount Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetLightSourceCount Method (IModelDoc2)

Gets the number of light sources.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLightSourceCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Integer

value = instance.GetLightSourceCount()
```

C#

```
System.int GetLightSourceCount()
```

C++/CLI

```
System.int GetLightSourceCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of light sources

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetLightSourceCount](#).

Example

[Turn Lights On \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetLightSourceExtProperty Method](#)
[IModelDoc2::GetLightSourceIdFromName Method](#)
[IModelDoc2::GetLightSourceName Method](#)
[IModelDoc2::SetLightSourceName Method](#)
[IModelDoc2::SetLightSourcePropertyValuesVB Method](#)
[IModelDoc2::ILightSourcePropertyValues Property](#)
[IModelDoc2::LightSourcePropertyValues Property](#)
[IModelDoc2::LightSourceUserName Property](#)
[IModelDoc2::AddLightSource Method](#)
[IModelDoc2::AddLightSourceExtProperty Method](#)
[IModelDoc2::DeleteLightSource Method](#)
[IModelDoc2::ResetLightSourceExtProperty Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetLightSourceExtProperty Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetLightSourceExtProperty Method (IModelDoc2)

Gets a float, string, or integer value stored for the light source.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLightSourceExtProperty( _  
    ByVal ID As System.Integer, _  
    ByVal PropertyId As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ID As System.Integer  
Dim PropertyId As System.Integer  
Dim value As System.Object
```

```
value = instance.GetLightSourceExtProperty(ID, PropertyId)
```

C#

```
System.object GetLightSourceExtProperty(  
    System.int ID,  
    System.int PropertyId  
)
```

C++/CLI

```
System.Object^ GetLightSourceExtProperty(  
& System.int ID,  
& System.int PropertyId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID

ID of the light source

PropertyId

ID of the property extension (see **Remarks**)

Return Value

Value stored for the light source

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetLightSourceExtProperty](#).

Remarks

The type returned is based on the how the data was placed. See [IModelDoc2::AddLightSourceExtProperty](#) for details.

Values for *Id* can range from 0 to *n*, where *n* = (the total number of light sources - 1). To get the total number of light sources, use [IModelDoc2::GetLightSourceCount](#).

The *PropertyId* parameter takes a 0-based index of the property. Values returned from [IModelDoc2::AddLightSourceExtProperty](#) must be decremented by 1 when used as input to [IModelDoc2::GetLightSourceExtProperty](#). However, do not decrement the error value of -1.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddLightSource Method](#)
[IModelDoc2::AddLightSourceExtProperty Method](#)
[IModelDoc2::DeleteLightSource Method](#)
[IModelDoc2::GetLightSourceCount Method](#)
[IModelDoc2::GetLightSourceIdFromName Method](#)
[IModelDoc2::GetLightSourceName Method](#)
[IModelDoc2::SetLightSourceName Method](#)
[IModelDoc2::SetLightSourcePropertyValuesVB Method](#)
[IModelDoc2::LightSourcePropertyValues Property](#)
[IModelDoc2::LightSourcePropertyValues Property](#)
[IModelDoc2::LightSourceUserName Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetLightSourceIdFromName Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : [GetLightSourceIdFromName Method \(IModelDoc2\)](#)

Gets the ID of the specified light source.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLightSourceIdFromName( _  
    ByVal LightName As System.String _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim LightName As System.String  
Dim value As System.Integer
```



```
value = instance.GetLightSourceIdFromName(LightName)
```

C#

```
System.int GetLightSourceIdFromName(  
    System.string LightName  
)
```

C++/CLI

```
System.int GetLightSourceIdFromName(  
& System.String^ LightName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

LightName
Internal name of the light source

Return Value

ID for the light source

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetLightSourceIdFromName](#).

Remarks

The ID for the light source can range from 0 to n , where n = (the total number of light sources - 1). To get the total number of light sources, use [IModelDoc2::GetLightSourceCount](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddLightSource Method](#)
[IModelDoc2::AddLightSourceExtProperty Method](#)
[IModelDoc2::DeleteLightSource Method](#)
[IModelDoc2::GetLightSourceCount Method](#)
[IModelDoc2::GetLightSourceExtProperty Method](#)
[IModelDoc2::GetLightSourceName Method](#)
[IModelDoc2::ResetLightSourceExtProperty Method](#)
[IModelDoc2::SetLightSourceName Method](#)
[IModelDoc2::SetLightSourcePropertyValuesVB Method](#)
[IModelDoc2::ILightSourcePropertyValues Property](#)
[IModelDoc2::LightSourcePropertyValues Property](#)
[IModelDoc2::LightSourceUserName Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetLightSourceName Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetLightSourceName Method (IModelDoc2)

Gets the name of a light source used internally by the SOLIDWORKS Design application.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLightSourceName(  
    ByVal ID As System.Integer  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim ID As System.Integer
Dim value As System.String

value = instance.GetLightSourceName(ID)
```

C#

```
System.string GetLightSourceName(
    System.int ID
)
```

C++/CLI

```
System.String^ GetLightSourceName(
    & System.int ID
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID
ID of light source

Return Value

Name of the light source

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetLightSourceName](#).

Example

[Turn Lights On \(VBA\)](#)

Remarks

Values for id can range from 0 to n , where n = (the total number of light sources - 1). To get the total number of light sources, use [IModelDoc2::GetLightSourceCount](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddLightSource Method](#)
[IModelDoc2::AddLightSourceExtProperty Method](#)
[IModelDoc2::DeleteLightSource Method](#)
[IModelDoc2::GetLightSourceCount Method](#)
[IModelDoc2::GetLightSourceExtProperty Method](#)
[IModelDoc2::GetLightSourceIdFromName Method](#)
[IModelDoc2::GetLightSourceName Method](#)
[IModelDoc2::ResetLightSourceExtProperty Method](#)
[IModelDoc2::SetLightSourceName Method](#)
[IModelDoc2::SetLightSourcePropertyValuesVB Method](#)
[IModelDoc2::ILightSourcePropertyValues Property](#)
[IModelDoc2::LightSourcePropertyValues Property](#)
[IModelDoc2::LightSourceUserName Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetLineCount Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetLineCount Method (IModelDoc2)

Gets the number of lines in the current sketch.

.NET Syntax

```
Function GetLineCount() As System.Integer
```



```
Dim instance As IModelDoc2
Dim value As System.Integer

value = instance.GetLineCount()
```

```
System.int GetLineCount()
```

```
System.int GetLineCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of lines in the current sketch

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetLineCount](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetLines Method](#)
[IModelDoc2::IGetLines Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetLines Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetLines Method (IModelDoc2)

Gets all of the lines in the current sketch.

.NET Syntax

```
Function GetLines() As System.Object
```



```
Dim instance As IModelDoc2
Dim value As System.Object

value = instance.GetLines()
```

```
System.object GetLines()
```

```
System.Object^ GetLines();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetLines](#).

Remarks

The return value is the following array of doubles:

[*LineType*, *StartPtX*, *StartPtY*, *StartPtZ*, *EndPtX*, *EndPtY*, *EndPtZ*, ...]

where this array of 7 values repeats itself for each line in the current sketch. The number of doubles returned is (*lineCount* * 7). To determine the number of lines in the current sketch, use [IModelDoc2::GetLineCount](#).

See [swLineTypes_e](#) for valid line types.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IGetLines Method](#)
[IModelDoc2::GetLineCount Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetMassProperties Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetMassProperties Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::GetMassProperties](#) and [IModelDocExtension::IGetMassProperties](#).

.NET Syntax

Visual Basic (Declaration)

Function GetMassProperties() As System.Object

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Object  
  
value = instance.GetMassProperties()
```

C#

```
System.Object GetMassProperties()
```

C++/CLI

```
System.Object^ GetMassProperties();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetMassProperties](#).

See Also

GetMassProperties2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetMassProperties2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::GetMassProperties](#) and [IModelDocExtension::IGetMassProperties](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetMassProperties2( _  
    ByRef Status As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Status As System.Integer  
Dim value As System.Object  
  
value = instance.GetMassProperties2(Status)
```

C#

```
System.object GetMassProperties2(  
    out System.int Status  
)
```

C++/CLI

```
System.Object^ GetMassProperties2(  
& [Out] System.int Status  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Status

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetMassProperties2](#).

See Also

GetModelViewCount Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetModelViewCount Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::GetModelViewCount](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetModelViewCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Integer

value = instance.GetModelViewCount()
```

C#

```
System.int GetModelViewCount()
```

C++/CLI

```
System.int GetModelViewCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of [model views](#) in this document

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetModelViewCount](#).

Example

[Get View Mode Names for Model \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::EnumModelViews Method](#)
[IModelDoc2::GetFirstModelView Method](#)
[IModelDoc2::GetModelViewNames Method](#)
[IModelDoc2::IGetModelViewNames Method](#)
[IModelDoc2::ActiveView Property](#)
[IModelDoc2::IActiveView Property](#)
[IModelDoc2::ModelViewManager Property](#)
[IModelDoc2::IGetFirstModelView Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetModelViewNames Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetModelViewNames Method (IModelDoc2)

Gets a list containing the names of each model view in this document.

.NET Syntax

Visual Basic (Declaration)

```
Function GetModelViewNames() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Object

value = instance.GetModelViewNames()
```

C#

```
System.object GetModelViewNames()
```

C++/CLI

```
System.Object^ GetModelViewNames();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array containing the names of each model view in this document

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetModelViewNames](#).

Example

[Get View Mode Names for Model \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ModelViewManager Property](#)
[IModelDoc2::ActiveView Property](#)
[IModelDoc2::ActiveView Property](#)
[IModelDoc2::GetModelViewNames Method](#)
[IModelDoc2::GetModelViewCount Method](#)
[IModelDoc2::GetFirstModelView Method](#)
[IModelDoc2::GetFirstModelView Method](#)
[IModelDoc2::EnumModelViews Method](#)
[IModelView Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetNext Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetNext Method (IModelDoc2)

Gets the next document opened in the current SOLIDWORKS Design session.

.NET Syntax

Visual Basic (Declaration)

```
Function GetNext() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Object  
  
value = instance.GetNext()
```

C#

```
System.object GetNext()
```

C++/CLI

```
System.Object^ GetNext();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Next [model document](#) opened in the current SOLIDWORKS Design session

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetNext](#).

Example

[Determine If Document Is Dirty \(VBA\)](#)
[Get Names of Open Documents \(VBA\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(C#\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(VB.NET\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(VBA\)](#)

Remarks

To traverse all documents open in this SOLIDWORKS Design session, [ISldWorks::GetFirstDocument](#) must have returned the first IModelDoc2 object that calls this method.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IGetNext Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetNumDependencies Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetNumDependencies Method (IModelDoc2)

Gets the number of strings returned by [IModelDoc2::GetDependencies2](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetNumDependencies( _  
    ByVal Traverseflag As System.Integer, _  
    ByVal Searchflag As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Traverseflag As System.Integer  
Dim Searchflag As System.Integer  
Dim value As System.Integer  
  
value = instance.GetNumDependencies(Traverseflag, Searchflag)
```

C#

```
System.int GetNumDependencies(  
    System.int Traverseflag,  
    System.int Searchflag  
)
```

C++/CLI

```
System.int GetNumDependencies(  
& System.int Traverseflag,  
& System.int Searchflag  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Traverseflag

True to traverse down into all dependent files, false for only the highest level within the dependencies

Searchflag

True to apply the current search criteria, false to return the dependent file information as it was stored

Return Value

Number of strings returned by [IModelDoc2::GetDependencies2](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetNumDependencies](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetNumDependencies2 Method](#)
[IModelDoc2::GetDependencies2 Method](#)
[ISldWorks::GetDocumentDependencies2 Method](#)
[ISldWorks::GetDocumentDependenciesCount Method](#)
[ISldWorks::GetDocumentDependencies2 Method](#)
[ISldWorks::GetDocumentDependenciesCount2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPathName Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetPathName Method (IModelDoc2)

Gets the full path name for this document, including the file name.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPathName() As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.String  
  
value = instance.GetPathName()
```

C#

```
System.string GetPathName()
```

C++/CLI

```
System.String^ GetPathName();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Full path name for this document including the file name; if the document has not yet been saved, then the string is empty

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetPathName](#).

Example

[Get List of Open Documents \(C++\)](#)
[Find Total Length of Sketch Segments in Sketch \(VBA\)](#)
[Get Active Document Dependents \(VBA\)](#)
[Get File Summary Information \(VBA\)](#)
[Get Unopened Document Dependents \(VBA\)](#)
[Get Paths of Open Documents \(C#\)](#)
[Get Paths of Open Documents \(VB.NET\)](#)
[Get Paths of Open Documents \(VBA\)](#)
[Get List of Configurations \(C#\)](#)

[Get List of Configurations \(VB.NET\)](#)
[Get List of Configurations \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IComponent2::GetPathName Method \(\)](#)
[IComponent2::GetImportedPath Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetPointLightProperties Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetPointLightProperties Method (IModelDoc2)

Gets point light properties.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPointLightProperties( _  
    ByVal Name As System.String, _  
    ByRef Ambient As System.Double, _  
    ByRef Diffuse As System.Double, _  
    ByRef Specular As System.Double, _  
    ByRef Colour As System.Integer, _  
    ByRef Enabled As System.Boolean, _  
    ByRef Fixed As System.Boolean, _  
    ByRef X As System.Double, _  
    ByRef Y As System.Double, _  
    ByRef Z As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Name As System.String  
Dim Ambient As System.Double  
Dim Diffuse As System.Double  
Dim Specular As System.Double  
Dim Colour As System.Integer  
Dim Enabled As System.Boolean  
Dim Fixed As System.Boolean  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Boolean
```

```
value = instance.GetPointLightProperties(Name, Ambient, Diffuse, Specular, Colour, Enabled, Fixed,  
X, Y, Z)
```

C#

```
System.bool GetPointLightProperties(  
    System.string Name,  
    ref System.double Ambient,  
    ref System.double Diffuse,  
    ref System.double Specular,  
    ref System.int Colour,  
    ref System.bool Enabled,  
    ref System.bool Fixed,  
    ref System.double X,  
    ref System.double Y,
```

```
        ref System.double Z
    )
```

C++/CLI

```
System.bool GetPointLightProperties(
& System.String^ Name,
& System.double% Ambient,
& System.double% Diffuse,
& System.double% Specular,
& System.int% Colour,
& System.bool% Enabled,
& System.bool% Fixed,
& System.double% X,
& System.double% Y,
& System.double% Z
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Light name

Ambient

Light source ambient value

Diffuse

Light source diffuse value

Specular

Light source specular value

Colour

COLORREF color value

Enabled

True if a light is enabled, false if not

Fixed

True if a light is fixed, false if not

X

x location of the point light

Y

y location of the point light

Z

z location of the point li

Return Value

True if light properties determined without a problem, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetPointLightProperties](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::SetPointLightProperties Method](#)

[IModelDoc2::GetAmbientLightProperties Method](#)

[IModelDoc2::GetDirectionLightProperties Method](#)

[IModelDoc2::GetSpotlightProperties Method](#)

[IModelDoc2::SetAmbientLightProperties Method](#)

[IModelDoc2::SetDirectionLightProperties Method](#)

[IModelDoc2::SetSpotlightProperties Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPopupMenuMode Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetPopupMenuMode Method (IModelDoc2)

Gets the current pop-up menu mode.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPopupMenuMode() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Integer

value = instance.GetPopupMenuMode()
```

C#

```
System.int GetPopupMenuMode()
```

C++/CLI

```
System.int GetPopupMenuMode();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Current pop-up menu mode:

- 0 - Default shortcut mode. This presents end users with options to **Select Other**, manipulate the view, access the properties dialog of the selected item, and so on.
- 1 - End users are presented with a limited set of choices including **Select Other** and **Clear Selection**. This mode is typically seen when a SOLIDWORKS dialog is active and the end user is restricted to entity selection.

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetPopupMenuMode](#).

Remarks

When end users click the right-mouse button when the pointer is on an entity in the graphics window, they are presented with one of two distinct menu sets. These menu sets have been defined as mode 0 and mode 1.

Using [IModelDoc2::SetPopupMenuMode](#), your application can simulate the same shortcut menu behavior as in the SOLIDWORKS Design user interface. If you have a dialog that requires end-user selection of entities, you can set the pop-up menu mode to 1 to simulate SOLIDWORKS behavior. Your application should always set the menu mode back to its previous value. You can determine the previous value by calling the [IModelDoc2::GetPopupMenuMode](#) before calling [IModelDoc2::SetPopupMenuMode](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPropertyExtension Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetPropertyExtension Method (IModelDoc2)

Gets the specified property extension on this model.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPropertyExtension( _  
    ByVal ID As System.Integer _
```

```
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim ID As System.Integer
Dim value As System.Object

value = instance.GetPropertyExtension(ID)
```

C#

```
System.object GetPropertyExtension(
    System.int ID
)
```

C++/CLI

```
System.Object^ GetPropertyExtension(
    & System.int ID
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID

(Size of the array returned by [IModelDoc2::AddPropertyExtension](#)) - (Position of the property extension in the array)

Return Value

Value of the property extension

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetPropertyExtension](#).

Remarks

IModelDoc2::AddPropertyExtension adds property extensions to a last-in-first-out, 1-based array and returns the size of that array.

NOTE: SOLIDWORKS recommends that you use the [IAttribute](#), [IAttributeDef](#), and [IParameter](#) interfaces instead of this method. These interface provide more flexibility.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ResetPropertyExtension Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetPropertyManagerPage Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetPropertyManagerPage Method (IModelDoc2)

Obsolete. Superseded by [ISldWorks::CreatePropertyManagerPage](#) and [ISldWorks::ICreatePropertyManagerPage](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetPropertyManagerPage( _
    ByVal DialogId As System.Integer, _
    ByVal Title As System.String, _
```

```
ByVal Handler As System.Object _  
) As PropertyManagerPage
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim DialogId As System.Integer  
Dim Title As System.String  
Dim Handler As System.Object  
Dim value As PropertyManagerPage  
  
value = instance.GetPropertyManagerPage(DialogId, Title, Handler)
```

C#

```
PropertyManagerPage GetPropertyManagerPage(  
    System.int DialogId,  
    System.string Title,  
    System.object Handler  
)
```

C++/CLI

```
PropertyManagerPage^ GetPropertyManagerPage(  
& System.int DialogId,  
& System.String^ Title,  
& System.Object^ Handler  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DialogId

Title

Handler

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetPropertyManagerPage](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetRayIntersectionsPoints Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetRayIntersectionsPoints Method (IModelDoc2)

Gets the intersection point information generated by [IModelDoc2::RayIntersections](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetRayIntersectionsPoints() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Object  
  
value = instance.GetRayIntersectionsPoints()
```

C#

System.Object GetRayIntersectionsPoints()

C++/CLI

System.Object^ GetRayIntersectionsPoints();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array containing the hit points (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetRayIntersectionsPoints](#).

Remarks

The array returned contains information about each hit of each ray on the given bodies, in the following format:

For each intersection found:

- (double) BodyIndex: index in BodiesIn array of the body that was hit
- (double) RayIndex: index in VectorsIn array of rays
- (double) IntersectionType: type of intersection found. See [swRayPtsResults_e](#) . Entry/Exit information (swRayPtsResultsENTER and swRayPtsResultsEXIT) are only returned when swRayPtsOptsENTRY_EXIT is specified in the options argument in the [IModelDoc2::RayIntersections](#) call. If more than one intersection type is found, then the IntersectionType contains an OR'd set of values from the swRayPtsResults_e enumeration.
- (double) x,y,z: coordinates of hit point
- (double) nx,ny,nz: normal (optionally appears when swRayPtsOptsNORMALS is specified in the Options argument in the IModelDoc2::RayIntersections call).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IGetRayIntersectionsPoints Method](#)
[IModelDoc2::IRayIntersections Method](#)
[IModelDoc2::IGetRayIntersectionsTopology Method](#)
[IModelDoc2::GetRayIntersectionsTopology Method](#)
[IModelDoc2::MultiSelectByRay Method](#)
[IModelDoc2::IMultiSelectByRay Method](#)
[IModelDoc2::ISelectByRay Method](#)
[IModelDoc2::SelectByRay Method](#)
[IModelDoc2::RayIntersections Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetRayIntersectionsTopology Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetRayIntersectionsTopology Method (IModelDoc2)

Gets the topology intersections generated by [IModelDoc2::RayIntersections](#).

.NET Syntax

Visual Basic (Declaration)

Function GetRayIntersectionsTopology() As System.Object

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Object

value = instance.GetRayIntersectionsTopology()
```

C#

```
System.Object GetRayIntersectionsTopology()
```

C++/CLI

```
System.Object^ GetRayIntersectionsTopology();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of containing the hit points (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetRayIntersectionsTopology](#).

Remarks

This method only returns information if the swRayPtsOptsTOPOLS option was specified in the Options argument of [IModelDoc2::RayIntersections](#).

The array returned contains information about each entity hit of each ray on the given bodies. The topology hit can be coordinated with the hit point using the BodyIndex return value from [IModelDoc2::GetRayIntersectionsPoints](#). You can determine the type of item hit (face, edge or vertex) by checking the IntersectionType return value in the IModelDoc2::GetRayIntersectionsPoints.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IGetRayIntersectionsTopology Method](#)
[IModelDoc2::IGetRayIntersectionsPoints Method](#)
[IModelDoc2::MultiSelectByRay Method](#)
[IModelDoc2::IMultiSelectByRay Method](#)
[IModelDoc2::IRayIntersections Method](#)
[IModelDoc2::SelectByRay Method](#)
[IModelDoc2::ISelectByRay Method](#)
[IModelDoc2::RayIntersections Method](#)
[IModelDoc2::GetRayIntersectionsPoints Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSaveFlag Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetSaveFlag Method (IModelDoc2)

Gets whether the document is currently dirty and needs to be saved.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSaveFlag() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Boolean

value = instance.GetSaveFlag()
```

C#

```
System.bool GetSaveFlag()
```

C++/CLI

```
System.bool GetSaveFlag();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this document needs to be saved, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetSaveFlag](#).

Example

[Determine If Document Is Dirty \(VBA\)](#)

Remarks

This flag:

- determines if the **Do you wish to save changes?** dialog is displayed when the user tries to close the document. Many operations cause this flag to be set, and you can use [IModelDoc2::SetSaveFlag](#) to set this flag.
- is set to true if the document was created in an older version of SOLIDWORKS.
- is set to true for assemblies only when a subassembly has been saved. If this flag is set to true for a subassembly, then the assembly is not marked as dirty until the subassembly is saved.
- is reset to false after an end-user has saved the file.

NOTE: This method returns true if the model is opened read-only, the system option **Don't prompt to save read-only referenced documents** is not selected, and the model is dirty and visible.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SetSaveFlag Method](#)
[IConfiguration::IsDirty Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSceneBkgDIB Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetSceneBkgDIB Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::GetSceneBkgDIBx64](#).

This method is unsafe to use with 64-bit SOLIDWORKS, i.e. all versions since SOLIDWORKS 2016.

The 32-bit integer returned may be too small to contain the 64-bit window handle/pointer, and will be truncated silently. A critical exception will occur sometime later, when the truncated handle/pointer is used.

Gets background image as a LPDIBSECTION.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSceneBkgDIB() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Integer
```

```
value = instance.GetSceneBkgDIB()
```

C#

```
System.int GetSceneBkgDIB()
```

C++/CLI

```
System.int GetSceneBkgDIB();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Background image as DIBSECTION

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetSceneBkgDIB](#).

Remarks

All applications are required to be compatible with SOLIDWORKS x64. You must replace all calls to 32-bit only APIs with calls to their 64-bit safe counterparts.

See [Eliminating 32-bit only API use](#) for more information.

The memory for the image bits (DIBSECTION.dsBm.bmBits) and this structure are allocated by the SOLIDWORKS Design application and must be deleted by the caller.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SetSceneBkgDIB Method](#)
[IModelDocExtension::SetSceneBkgDIBx64 Method](#)
[IModelDoc2::SceneBkgImageFileName Property](#)
[IModelDoc2::InsertBkgImage Method](#)
[IModelDoc2::DeleteBkgImage Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSceneExtProperty Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetSceneExtProperty Method (IModelDoc2)

Gets a float, string, or integer value stored for the scene.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSceneExtProperty( _  
    ByVal PropertyId As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim PropertyId As System.Integer  
Dim value As System.Object  
  
value = instance.GetSceneExtProperty(PropertyId)
```

C#

```
System.object GetSceneExtProperty(  
    System.int PropertyId  
)
```

C++/CLI

```
System.Object^ GetSceneExtProperty(  
& System.int PropertyId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PropertyId
ID of the property extension

Return Value

Value stored for the scene extension property

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetSceneExtProperty](#).

Remarks

The type returned is based on the how the data was placed. See [ModelDoc2::AddSceneExtProperty](#) for details.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ResetSceneExtProperty Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSpotlightProperties Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetSpotlightProperties Method (IModelDoc2)

Gets the spotlight properties.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSpotlightProperties( _  
    ByVal Name As System.String, _  
    ByRef Ambient As System.Double, _  
    ByRef Diffuse As System.Double, _  
    ByRef Specular As System.Double, _  
    ByRef Colour As System.Integer, _  
    ByRef Enabled As System.Boolean, _  
    ByRef Fixed As System.Boolean, _  
    ByRef X As System.Double, _  
    ByRef Y As System.Double, _  
    ByRef Z As System.Double, _  
    ByRef Targetx As System.Double, _  
    ByRef Targety As System.Double, _  
    ByRef Targetz As System.Double, _  
    ByRef ConeAngle As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim Name As System.String
Dim Ambient As System.Double
Dim Diffuse As System.Double
Dim Specular As System.Double
Dim Colour As System.Integer
Dim Enabled As System.Boolean
Dim Fixed As System.Boolean
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim Targetx As System.Double
Dim Targety As System.Double
Dim Targetz As System.Double
Dim ConeAngle As System.Double
Dim value As System.Boolean

```

```

value = instance.GetSpotlightProperties(Name, Ambient, Diffuse, Specular, Colour, Enabled, Fixed,
X, Y, Z, Targetx, Targety, Targetz, ConeAngle)

```

C#

```

System.bool GetSpotlightProperties(
    System.string Name,
    ref System.double Ambient,
    ref System.double Diffuse,
    ref System.double Specular,
    ref System.int Colour,
    ref System.bool Enabled,
    ref System.bool Fixed,
    ref System.double X,
    ref System.double Y,
    ref System.double Z,
    ref System.double Targetx,
    ref System.double Targety,
    ref System.double Targetz,
    ref System.double ConeAngle
)

```

C++/CLI

```

System.bool GetSpotlightProperties(
    & System.String^ Name,
    & System.double% Ambient,
    & System.double% Diffuse,
    & System.double% Specular,
    & System.int% Colour,
    & System.bool% Enabled,
    & System.bool% Fixed,
    & System.double% X,
    & System.double% Y,
    & System.double% Z,
    & System.double% Targetx,
    & System.double% Targety,
    & System.double% Targetz,
    & System.double% ConeAngle
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name
Light name used internally (returned by [IModelDoc2::GetLightSourceName](#))

Ambient
Light source ambient value

Diffuse
Light source diffuse value

Specular
Light source specular value

Colour

COLORREF color value

Enabled

True if a light is enabled, false if not

Fixed

True if a light is fixed, false if not

X

x location of the spot light

Y

y location of the spot light

Z

location of the spot light target

Targetx

x location of the spot light target

Targety

y location of the spot light target

Targetz

z location of the spot light target

ConeAngle

Cone angle through which the beam spreads in degrees

Return Value

True if light properties determined without a problem, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetSpotlightProperties](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::SetPointLightProperties Method](#)

[IModelDoc2::GetAmbientLightProperties Method](#)

[IModelDoc2::GetDirectionLightProperties Method](#)

[IModelDoc2::GetPointLightProperties Method](#)

[IModelDoc2::SetAmbientLightProperties Method](#)

[IModelDoc2::SetDirectionLightProperties Method](#)

[IModelDoc2::SetPointLightProperties Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetStandardViewRotation Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetStandardViewRotation Method (IModelDoc2)

Gets the specified view orientation matrix with respect to the Front view.

.NET Syntax

Visual Basic (Declaration)

```
Function GetStandardViewRotation( _  
    ByVal ViewId As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ViewId As System.Integer  
Dim value As System.Object
```

```
value = instance.GetStandardViewRotation(ViewId)
```

C#

```
System.Object GetStandardViewRotation(  
    System.int ViewId
```

)

C++/CLI

```
System.Object^ GetStandardViewRotation(  
& System.int ViewId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ViewId

View ID that you want to evaluate as defined in [swStandardViews_e](#)

Return Value

Array describing the view rotation with respect to the Front view; this is an array of 9 doubles

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetStandardViewRotation](#).

Example

[Show Named View \(VBA\)](#)

Remarks

The end-user may have redefined the Front view to be something other than the X-Y plane.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IGetStandardViewRotation Method](#)
[IModelDocExtension::IGetNamedViewRotation Method](#)
[IModelDocExtension::GetNamedViewRotation Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetTessellationQuality Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetTessellationQuality Method (IModelDoc2)

Gets the shaded-display image quality number for the current document.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessellationQuality() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Integer  
  
value = instance.GetTessellationQuality()
```

C#

```
System.int GetTessellationQuality()
```

C++/CLI

```
System.int GetTessellationQuality();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

0 < Shaded-display image quality number < 106 (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetTessellationQuality](#).

Remarks

This method returns a number (*QualityNum*) that corresponds to the degree of tessellation of curved surfaces for shaded rendering output. **Tool** > **Options** > **Document Properties** > **Image Quality** includes a field that sets the maximum chordal deviation. *QualityNum* and the maximum chordal deviation (in meters) are coupled and inversely proportional as follows:

$$var = TessMin + QualityNum * ((TessMax - TessMin) / 100)$$

$$Deviation = 0.025 * (BodyDiameter * 2) / var;$$

where:

- *TessMin* = 6
- *TessMax* = 166
- 0 < *QualityNum* (returned by this method) < 106
- *BodyDiameter* is the diagonal distance across the bounds of the part box. See [IPartDoc::GetPartBox](#) or [IPartDoc::IGetPartBox](#) for more information.

You can also get the maximum chordal deviation using [swUserPreferencesDouble_e.swImageQualityShadedDeviation](#).

To get the image quality number, use [swUserPreferenceInteger_e.swImageQualityShaded](#).

NOTE: [Setting the degree of tessellation](#) to a higher value results in:

- finer tessellation,
- increased file size,
- slower graphics performance, and
- increased memory usage.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTitle Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetTitle Method (IModelDoc2)

Gets the title of the document that appears in the active window's title bar.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTitle() As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.String
```

```
value = instance.GetTitle()
```

C#

```
System.string GetTitle()
```

C++/CLI

```
System.String^ GetTitle();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Title string, usually displayed on the window's title bar

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetTitle](#).

Example

[Detecting In-context Edit \(C++\)](#)
[Close Document \(VBA\)](#)
[Get Document Information \(VBA\)](#)
[Get Names of Components and Window Handle and DIBSECTION \(VBA\)](#)
[Get Names of Open Documents \(VBA\)](#)
[Save Rollbacked Part As Parasolid File \(VBA\)](#)
[Traverse Bodies \(C++\)](#)
[Get Names of Configurations Using Variant \(C++\)](#)

Remarks

The document name that appears in the window header changes based on your File Explorer settings. If you chose to suppress known file extensions, then the title shown in the window, and returned by this method, varies (for example, **Part1.sldprt** vs. **Part1**)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SetTitle2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetToolBarVisibility Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetToolBarVisibility Method (IModelDoc2)

Gets the visibility of a toolbar.

.NET Syntax

Visual Basic (Declaration)

```
Function GetToolBarVisibility( _  
    ByVal ToolBar As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim Toolbar As System.Integer  
Dim value As System.Boolean
```

```
value = instance.GetToolBarVisibility(ToolBar)
```

C#

```
System.bool GetToolBarVisibility(  
    System.int ToolBar  
)
```

C++/CLI



Copy Code

```
System.bool GetToolbarVisibility(  
& System.int Toolbar  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Toolbar

Identifier of toolbar as defined in [swToolbar_e](#)

Return Value

True if the toolbar is visible, false if it is hidden

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetToolbarVisibility](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SetToolbarVisibility Method](#)
[IModelDoc2::Toolbars Method](#)
[ISldWorks::AddToolbar4 Method](#)
[ISldWorks::GetToolbarState2 Method](#)
[ISldWorks::GetToolbarDock2 Method](#)
[ISldWorks::HideToolbar2 Method](#)
[ISldWorks::RemoveToolbar2 Method](#)
[ISldWorks::SetToolbarDock2 Method](#)
[ISldWorks::ShowToolbar2 Method](#)
[ICommandManager Interface](#)
[ICommandGroup Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetType Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetType Method (IModelDoc2)

Gets the type of the document.

.NET Syntax

Visual Basic (Declaration)

```
Function GetType() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Integer  
  
value = instance.GetType()
```

C#

```
System.int GetType()
```

C++/CLI

```
System.int GetType();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Type of this document as defined in [swDocumentTypes_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetType](#).

Example

[Display of Item in FeatureManager Design Tree \(C++\)](#)
[Get Edge Data By Name \(C++\)](#)
[Get Named Entities \(C++\)](#)
[Get Display Dimensions, GTols, and Surface-finish Symbols \(VBA\)](#)
[Get Material Properties \(VBA\)](#)
[Get Names of Bodies in Multibody Part \(VBA\)](#)
[Select Bodies \(VBA\)](#)
[Suppress or Unsuppress Feature \(VBA\)](#)
[Traverse Bodies \(C++\)](#)
[Get and Set Table Anchor of Hole Table \(C#\)](#)
[Get and Set Table Anchor of Hole Table \(VB.NET\)](#)
[Get and Set Table Anchor of Hole Table \(VBA\)](#)
[Get Names of Configurations Using Variant \(C++\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetUnits Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetUnits Method (IModelDoc2)

Gets the current unit settings, fraction base, fraction value, and significant digits.

.NET Syntax

Visual Basic (Declaration)

```
Function GetUnits() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Object

value = instance.GetUnits()
```

C#

```
System.object GetUnits()
```

C++/CLI

```
System.Object^ GetUnits();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of 5 shorts (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetUnits](#).

Remarks

The format of the returned data is an array of shorts:

[*LengthUnit*, *FractionBase*, *FractionValue*, *SignificantDigits*, *RoundToFraction*]

where:

- *LengthUnit* = current model units as defined in [swLengthUnit e.](#)
- *FractionBase* = decimal or fraction as defined in [swFractionDisplay e.](#); fractional units are only valid if the model units are set to swINCHES or swFEETINCHES.
- *FractionValue* = denominator value if using fractional units.
- *SignificantDigits* = significant digits if using decimal units.
- *RoundToFraction* = flag denoting whether or not to round to the fraction; for example, if 4 was the denominator value and the actual value was .27, it is rounded to .25

You can also use [IModelDoc2::LengthUnit](#), which provides access to the LengthUnit parameter.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IGetUnits Method](#)
[IModelDoc2::SetUnits Method](#)
[IModelDoc2::GetAngularUnits Method](#)
[IModelDoc2::GetUserUnit Method](#)
[IModelDoc2::IGetAngularUnits Method](#)
[IModelDoc2::IGetUserUnit Method](#)
[IModelDoc2::IGetAngularUnits Method](#)
[IModelDoc2::SetAngularUnits Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetUpdateStamp Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetUpdateStamp Method (IModelDoc2)

Gets the current update stamp for this document.

.NET Syntax

Visual Basic (Declaration)

```
Function GetUpdateStamp() As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim value As System.Integer

value = instance.GetUpdateStamp()
```

C#

```
System.int GetUpdateStamp()
```

C++/CLI

```
System.int GetUpdateStamp();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Current update stamp value for this document

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetUpdateStamp](#).

Remarks

The update stamp is an integer form of a time stamp. The update stamp is incremented for model state changes (i.e., suppress or unsuppress, configuration changes, feature changes, etc.) and for geometric changes (i.e., anything that requires a rebuild). This time stamp is not incremented for operations such as color changes, feature or configuration name changes, etc.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IFeature::GetUpdateStamp Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetUserPreferenceDoubleValue Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetUserPreferenceDoubleValue Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::GetUserPreferenceDouble](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetUserPreferenceDoubleValue( _  
    ByVal UserPreferenceValue As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim UserPreferenceValue As System.Integer  
Dim value As System.Double  
  
value = instance.GetUserPreferenceDoubleValue(UserPreferenceValue)
```

C#

```
System.double GetUserPreferenceDoubleValue(  
    System.int UserPreferenceValue  
)
```

C++/CLI

```
System.double GetUserPreferenceDoubleValue(  
& System.int UserPreferenceValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UserPreferenceValue

Value as defined in [swUserPreferenceDoubleValue_e](#)

Return Value

Numeric value of the user preference specified in UserPreferenceValue

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetUserPreferenceDoubleValue](#).

Example

[Get and Set Material Density \(VBA\)](#)
[Get Mass Properties of Assembly Component \(VBA\)](#)

Remarks

This method is equivalent to interactively getting document properties in the SOLIDWORKS Design software. See [System Options and Document Properties](#) for details.

If you want to retrieve the current, minimum, and maximum values for a document user preference, use [IModelDoc2::GetUserPreferenceDoubleValueRange](#). Be sure to read that method's **Remarks** section because its implementation is limited.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetUserPreferenceIntegerValue Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetUserPreferenceIntegerValue Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::GetUserPreferenceInteger](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetUserPreferenceIntegerValue( _  
    ByVal UserPreferenceValue As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim UserPreferenceValue As System.Integer  
Dim value As System.Integer  
  
value = instance.GetUserPreferenceIntegerValue(UserPreferenceValue)
```

C#

```
System.int GetUserPreferenceIntegerValue(  
    System.int UserPreferenceValue  
)
```

C++/CLI

```
System.int GetUserPreferenceIntegerValue(  
& System.int UserPreferenceValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UserPreferenceValue

Value as defined in [swUserPreferenceIntegerValue_e](#)

Return Value

Numeric value associated with the specified UserPreferenceValue

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetUserPreferenceIntegerValue](#).

Example

[Get Material Properties \(VBA\)](#)
[Make Part Transparent \(VBA\)](#)
[Save Document as TIFF \(VBA\)](#)
[Set Grid Lines \(VBA\)](#)

Remarks

This method is equivalent to interactively getting document properties in the SOLIDWORKS Design software. See [System Options and Document Properties](#) for details.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetUserPreferenceStringValue Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetUserPreferenceStringValue Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::GetUserPreferenceString](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetUserPreferenceStringValue( _  
    ByVal UserPreference As System.Integer _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim UserPreference As System.Integer  
Dim value As System.String  
  
value = instance.GetUserPreferenceStringValue(UserPreference)
```

C#

```
System.string GetUserPreferenceStringValue(  
    System.int UserPreference  
)
```

C++/CLI

```
System.String^ GetUserPreferenceStringValue(  
& System.int UserPreference  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UserPreference
Value as defined in [swUserPreferenceStringValue_e](#)

Return Value

String value of UserPreference

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetUserPreferenceStringValue](#).

Example

[Get Transform of Coordinate System \(VB.NET\)](#)
[Get Coordinate System Transform \(VBA\)](#)
[Save Rolledback Part as Parasolid File \(VBA\)](#)

Remarks

This method is equivalent to interactively getting document properties in the SOLIDWORKS Design software. See [System Options and Document Properties](#) for details.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetUserPreferenceTextFormat Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetUserPreferenceTextFormat Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::GetUserPreferenceTextFormat](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetUserPreferenceTextFormat( _  
    ByVal UserPreferenceValue As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim UserPreferenceValue As System.Integer  
Dim value As System.Object  
  
value = instance.GetUserPreferenceTextFormat(UserPreferenceValue)
```

C#

```
System.object GetUserPreferenceTextFormat(  
    System.int UserPreferenceValue  
)
```

C++/CLI

```
System.Object^ GetUserPreferenceTextFormat(  
& System.int UserPreferenceValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UserPreferenceValue
Value as defined in [swUserPreferenceTextFormat_e](#)

Return Value

[Text format](#) associated with UserPreferenceValue

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetUserPreferenceTextFormat](#).

Example

[Get Format of Note Text \(VBA\)](#)
[Replace Dimension With Text \(VBA\)](#)

Remarks

This method is equivalent to interactively getting document properties in the SOLIDWORKS Design software. See [System Options and Document Properties](#) for details.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetUserPreferenceToggle Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetUserPreferenceToggle Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::GetUserPreferenceToggle](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetUserPreferenceToggle( _  
    ByVal UserPreferenceToggle As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim UserPreferenceToggle As System.Integer  
Dim value As System.Boolean  
  
value = instance.GetUserPreferenceToggle(UserPreferenceToggle)
```

C#

```
System.bool GetUserPreferenceToggle(  
    System.int UserPreferenceToggle  
)
```

C++/CLI

```
System.bool GetUserPreferenceToggle(  
&    System.int UserPreferenceToggle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UserPreferenceToggle
Value as defined in [swUserPreferenceToggle_e](#)

Return Value

True if the item specified by UserPreferenceToggle is currently toggled on, false if the item is currently toggled off

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetUserPreferenceToggle](#).

Example

[Get Material Properties \(VBA\)](#)
[Ignore Feature Colors \(VBA\)](#)

Remarks

This method is equivalent to interactively getting document properties in the SOLIDWORKS Design software.

The value returned is true if the item is currently turned on, and false if the item is currently turned off. For example:

```
boolean curState = m_ModelDoc2.GetUserPreferenceToggle( swIgnoreFeatureColors )
```

See [System Options and Document Properties](#) for details.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetUserUnit Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetUserUnit Method (IModelDoc2)

Gets this document's units settings.

.NET Syntax

Visual Basic (Declaration)

```
Function GetUserUnit( _  
    ByVal UnitType As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim UnitType As System.Integer  
Dim value As System.Object  
  
value = instance.GetUserUnit(UnitType)
```

C#

```
System.object GetUserUnit(  
    System.int UnitType  
)
```

C++/CLI

```
System.Object^ GetUserUnit(  
& System.int UnitType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UnitType

Document unit type as defined in [swUserUnitsType_e](#)

Return Value

[IUserUnit](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetUserUnit](#).

Example

[Get and Set Document Units \(VBA\)](#)
[Get and Set Document Units \(VB.NET\)](#)
[Get and Set Document Units \(C#\)](#)

Remarks

The properties of IUserUnit are read only. Set them using the IModelDocExtension::SetUserPreference* methods.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetAngularUnits Method](#)
[IModelDoc2::GetUnits Method](#)
[IModelDoc2::IGetAngularUnits Method](#)
[IModelDoc2::IGetUnits Method](#)
[IModelDoc2::IGetUserUnit Method](#)
[IModelDoc2::ISetAngularUnits Method](#)
[IModelDoc2::SetUnits Method](#)
[IModelDoc2::SetAngularUnits Method](#)
[ISldWorks::GetUserUnit Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetVisibilityOfConstructPlanes Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetVisibilityOfConstructPlanes Method (IModelDoc2)

Gets whether construction (reference) planes are currently visible.

.NET Syntax

Visual Basic (Declaration)

```
Function GetVisibilityOfConstructPlanes() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Boolean

value = instance.GetVisibilityOfConstructPlanes()
```

C#

```
System.bool GetVisibilityOfConstructPlanes()
```

C++/CLI

```
System.bool GetVisibilityOfConstructPlanes();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if construction planes are currently visible, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetVisibilityOfConstructPlanes](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

[IModelDoc2::ViewDispRefplanes Method](#)
[IModelDoc2::CreatePlaneAtAngle3 Method](#)
[IModelDoc2::CreatePlaneAtOffset3 Method](#)
[IModelDoc2::CreatePlaneAtSurface3 Method](#)
[IModelDoc2::CreatePlaneFixed2 Method](#)
[IModelDoc2::CreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::CreatePlaneThru3Points3 Method](#)
[IModelDoc2::CreatePlaneThruLineAndPt Method](#)
[IModelDoc2::CreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::ICreatePlaneAtAngle3 Method](#)
[IModelDoc2::ICreatePlaneAtOffset3 Method](#)
[IModelDoc2::ICreatePlaneAtSurface3 Method](#)
[IModelDoc2::ICreatePlaneFixed2 Method](#)
[IModelDoc2::ICreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::ICreatePlaneThru3Points3 Method](#)
[IModelDoc2::ICreatePlaneThruLineAndPt Method](#)
[IModelDoc2::ICreatePlaneThruPtParallelToPlane Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetZebraStripeData Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GetZebraStripeData Method (IModelDoc2)

Gets zebra line data.

.NET Syntax

Visual Basic (Declaration)

```
Sub GetZebraStripeData( _  
    ByRef Size As System.Double, _  
    ByRef Ratio As System.Double, _  
    ByRef Color1 As System.Integer, _  
    ByRef Color2 As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Size As System.Double  
Dim Ratio As System.Double  
Dim Color1 As System.Integer  
Dim Color2 As System.Integer
```

```
instance.GetZebraStripeData(Size, Ratio, Color1, Color2)
```

C#

```
void GetZebraStripeData(  
    out System.double Size,  
    out System.double Ratio,  
    out System.int Color1,  
    out System.int Color2  
)
```

C++/CLI

```
void GetZebraStripeData(  
& [Out] System.double Size,  
& [Out] System.double Ratio,  
& [Out] System.int Color1,  
& [Out] System.int Color2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Size

Number of stripes

Ratio

Width of the stripes

Color1

First color in zebra stripe design

Color2

Second color in zebra stripe design

Return Value

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GetZebraStripeData](#).

Remarks

The size parameter is inversely related; a large size value generates lots of small stripes.

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::SetZebraStripeData Method](#)

[IModelView::DisplayZebraStripes Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GraphicsRedraw Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GraphicsRedraw Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::GraphicsRedraw2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub GraphicsRedraw()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.GraphicsRedraw()
```

C#

```
void GraphicsRedraw()
```

C++/CLI

```
void GraphicsRedraw();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GraphicsRedraw](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GraphicsRedraw2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GraphicsRedraw2 Method (IModelDoc2)

Obsolete. Superseded by [IModelView::GraphicsRedraw](#) and [IModelView::IGraphicsRedraw](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub GraphicsRedraw2()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.GraphicsRedraw2()
```

C#

```
void GraphicsRedraw2()
```

C++/CLI

```
void GraphicsRedraw2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GraphicsRedraw2](#).

Example

[Dynamic View Rotation \(C++\)](#)
[Insert Spline in Drawing \(C++\)](#)
[Autodimension a Sketch \(VBA\)](#)
[Insert DXF File and Add Dimension \(VBA\)](#)
[Redraw Graphics \(VBA\)](#)

Remarks

This method forces the display to be updated immediately when called.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::WindowRedraw Method](#)
[IModelView::DrawHighlightedItems Method](#)
[IModelView::DrawTransparentFeatureTree Method](#)
[IModelView::SuppressWaitCursorDuringRedraw Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0